
Cartographic Generalization Exposes Non-Uniform Structure Loss in Downsampled Dermoscopy Benchmarks

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Abstract

Downsampled medical imaging benchmarks ship a reduction with no name, no contract, and no statement of what it destroyed. Cartography solved the same problem two centuries ago with named generalization operators, legends, scale bars, and source diagrams. We implement that discipline for DermaMNIST and audit the benchmark as a data readiness question. The audit is decisive before any model is trained: four of five diagnostic structure families fall below the Nyquist limit of the shipped 28^2 grid, and the uniform resize retains structure non-uniformly across skin tone strata (spread 0.094, halved to 0.045 by equal-fidelity symbolization). We report costs in full: an 18 channel generalized sheet recovers 24% of the rare-class recall gap and improves calibration while losing 6.8 macro AUC points; a symbol budget scale law predicted by cartographic practice does not appear; and a label-free per-image certificate’s abstention value reverses between two runs of the same code. We propose four artifacts every such benchmark should ship, all machine-checkable today.

1 Introduction

Evaluations for scientific AI increasingly ask where a model is reliable rather than how it ranks. That question is unanswerable if the dataset itself cannot say where its own measurements are supported. MedMNIST-style benchmarks (Yang et al., 2023) reduce 224^2 clinical acquisitions to 28^2 by area resizing, an operation with no published contract: nothing in the release states which diagnostic structures survive the reduction, whether the loss is uniform across patients, or where an individual image stops supporting measurement at all. A mapmaker reducing 1:10,000 to 1:1,000,000 faces exactly this problem and never solved it by photoreduction. Cartographic generalization (Töpfer and Pillewizer, 1966; McMaster and Shea, 1992) names its operators, publishes what each preserves and sacrifices, prints a scale bar, and, on nautical charts, ships a source diagram showing where the survey cannot be trusted; the map is an argument about its data (Harley, 1989).

This paper applies that discipline to DermaMNIST and treats the benchmark as the object of study. Our contributions are a readiness audit that is a property of the data alone (the Nyquist audit and the source diagram), a controlled test of what an operator-based reduction buys and costs downstream, and an unusually complete account of what did not work, including a second run of the same code that moves several conclusions. Everything is reproducible from the released pipeline, whose fidelity

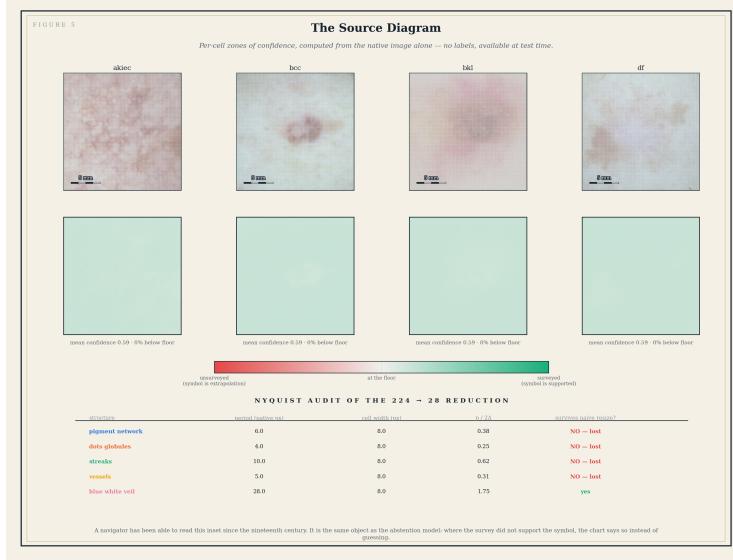


Figure 1: The source diagram: per-cell zones of confidence computed from the native image alone, with the Nyquist audit of Table 1. A chart has carried this inset since the nineteenth century; where the survey does not support the symbol, the sheet says so.

contracts are machine-readable and whose invariants are pinned by unit tests, a format compatible with machine-checkable review pilots such as this workshop’s.

2 Setup: three regimes, one protocol

DermaMNIST, the dermoscopy member of MedMNIST v2 (Yang et al., 2023), packages the HAM10000 archive (Tschandl et al., 2018) as 28×28 images produced by area resizing the 224^2 source. We treat that reduction as the object of study and compare three regimes trained and evaluated under one protocol. The **naïve** arm is the benchmark as shipped: the area-resized RGB image, which by construction equals channels 0 to 2 of our generalized tensor, so the two low resolution arms differ only in the presence of the added channels and never in interpolation kernel, colour round trip, or crop. The **generalized** arm is an 18 channel 28^2 sheet built from the same 224^2 source by cartographic generalization (Töpfer and Pillewizer, 1966; McMaster and Shea, 1992): 3 RGB channels; 12 symbol channels covering five dermoscopic structure families from the 7-point checklist (Argenziano et al., 1998) (a density channel per family, orientation encoded as $\cos 2\theta$ and $\sin 2\theta$ for the three oriented families with an equivariant augmenter, and a scale channel for dots and globules); and 3 source diagram channels (a noise margin, a quantization margin, and their minimum, the certificate), computed from the native image alone with no access to labels. The **native** 224^2 arm trains a ResNet-18 (He et al., 2016) from scratch on the source images as the pixel upper bound.

Each generalization operator (selection, simplification, aggregation, displacement, exaggeration, symbolization) carries a published fidelity contract stating what it preserves and sacrifices, and each contract names an invariant pinned by a unit test: aggregation conserves the areal integral, exaggeration preserves rank, simplification zeroes exactly below the coherence floor, and orientation pools on the projective line. Symbol normalization is anchored to a measured per-image noise floor, so a measurement the image cannot support collapses toward zero instead of amplifying noise into a confident false structure. Both low resolution arms train the same compact three-stage CNN (GroupNorm, SiLU, width 48); the only architectural difference is the stem’s input channel count. Training uses class-weighted cross entropy; probabilities are temperature scaled on validation only (Guo et al., 2017), and ECE is reported before and after scaling. Results in Sections 3 and 4 come from the released pipeline’s reduced budget configuration (per-class training counts appear in Figure 2; classes akiec, df, and vasc are rare in training). Section 5 reports a second, independently executed run on the full training split with three seeds.

Table 1: Nyquist audit of the $224 \rightarrow 28$ reduction. $\Delta = 8$ native pixels per benchmark cell; a structure survives resampling only if its period exceeds 2Δ .

structure	period (native px)	cell width (px)	$p/2\Delta$	survives naive resize?
pigment network	6.0	8.0	0.38	no, lost
dots and globules	4.0	8.0	0.25	no, lost
streaks	10.0	8.0	0.62	no, lost
vessels	5.0	8.0	0.31	no, lost
blue-white veil	28.0	8.0	1.75	yes

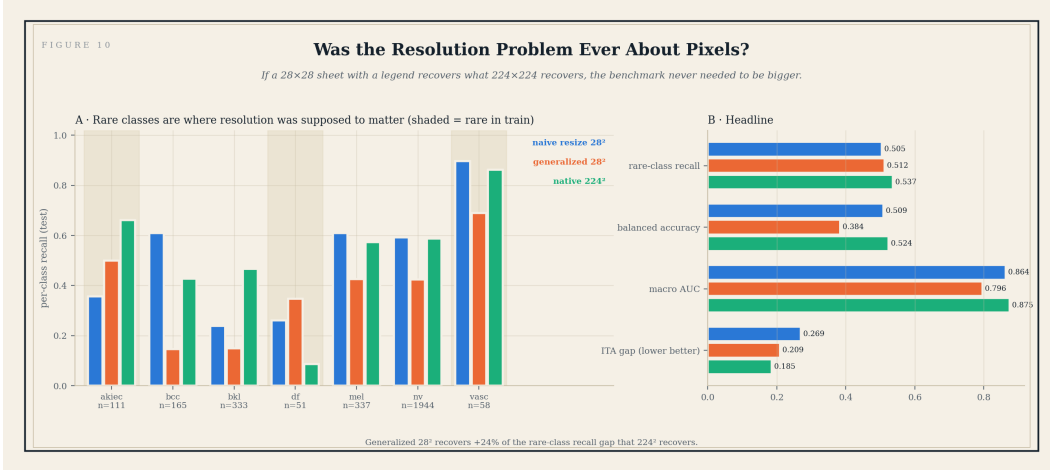


Figure 2: Per-class recall for the three regimes (A) and the headline metrics (B), reduced budget run. Shaded classes are rare in training; per-class training counts are printed beneath the class names. The generalized 28^2 sheet recovers 24% of the rare-class recall gap that 224^2 recovers, while several common classes lose recall.

3 The audit: what the 28^2 grid can carry at all

At $224 \rightarrow 28$ each benchmark cell spans $\Delta = 8$ native pixels, so any structure with spatial period below $2\Delta = 16$ native pixels is below the Nyquist limit of the target grid (Shannon, 1949): no resampling kernel at any quality setting can represent it. Table 1 gives the audit. Four of the five diagnostic families are unrepresentable on the shipped grid; symbolization does not beat the sampling theorem, it moves the measurement upstream of the reduction, which is a different operation with a different contract.

4 Results, including the costs

Aggregate quality conceals per-class behaviour. Table 2 and Figure 2 give the headline. The shipped benchmark looks healthy in aggregate (macro AUC 0.864, within 1.1 points of the 224^2 upper bound) while balanced accuracy sits at 0.509 and individual classes behave very differently: panel A of Figure 2 shows bcc recall dropping by roughly 0.45 between the naive and generalized arms, bkl recall below 0.25 in both low resolution arms, and the native 224^2 arm scoring lowest of all three arms on df. Resolution was supposed to matter most for rare classes, yet the shipped resize gives up only 3.2 points of rare recall against the 224^2 bound on this run, and the generalized sheet recovers 24% of that gap (0.505 to 0.512, against 0.537).

The generalized sheet pays for its properties. The 18 channel arm improves rare recall, calibration (ECE 0.062 versus 0.070), and the ITA gap (0.209 versus 0.269), and it costs 6.8 points of macro AUC and 12.5 points of balanced accuracy against the naive arm. We report this cost as a primary result rather than an implementation detail: on this dataset and budget, symbolization buys auditability and equity properties with aggregate accuracy.

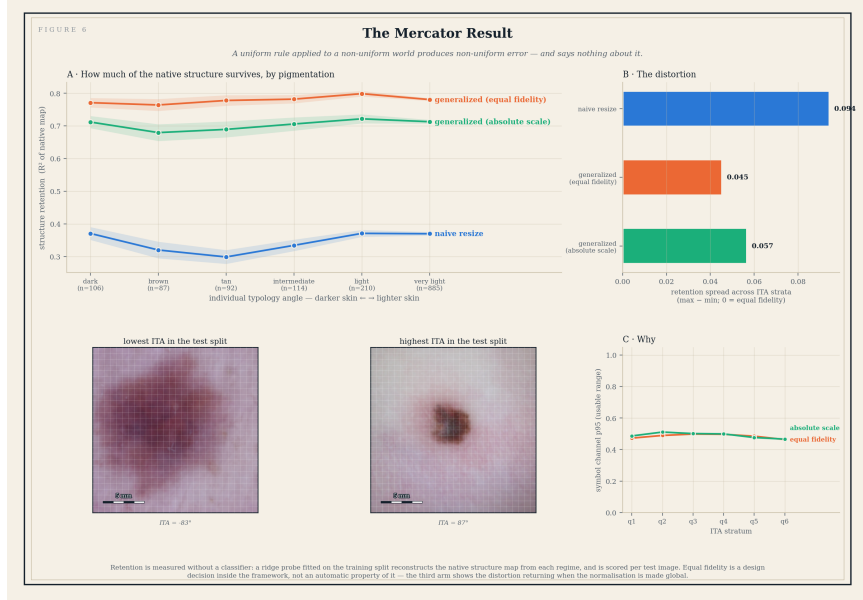


Figure 3: The retention result. A ridge probe reconstructs each structure family’s native density map from what a regime kept, scored per test image and stratified by ITA. Naive resizing retains least where pigmentation is darkest through tan and spreads 0.094 across strata; equal-fidelity generalization spreads 0.045; a global absolute scale, 0.057.

Table 2: Benchmark table for the reduced budget run (the run all figures are drawn from). ECE is post-temperature; the pre-temperature values are 0.071, 0.076, and 0.061 respectively. ITA gap is the largest between-stratum difference in balanced error across individual typology angle strata (lower is better).

regime	macro AUC	balanced acc.	rare recall	ECE	ITA gap
naive resize 28 ²	0.864	0.509	0.505	0.070	0.269
generalized 28 ²	0.796	0.384	0.512	0.062	0.209
native 224 ²	0.875	0.524	0.537	0.060	0.185

A uniform resize is a non-uniform measurement policy. Structure amplitude in L^* is coupled to constitutive pigmentation, so one low-pass rule loses different amounts of structure for different skin tones. We measure retention without any classifier: a ridge probe fitted on the training split reconstructs each family’s native density map from what a regime kept, scored per test image and stratified by individual typology angle (Del Bino and Bernerd, 2013; Kinyanjui et al., 2020) over the 1,494 test images with a finite ITA estimate (the strata are heavily skewed light: 885 of 1,494 fall in the lightest bin). Figure 3 shows the result: naive retention sits near 0.30 to 0.37 with its minimum in the tan stratum and a max minus min spread of 0.094 across strata; the generalized sheet with equal-fidelity normalization retains 0.77 to 0.80 with spread 0.045. A third arm replacing the per-image normalization with one global absolute scale lands at spread 0.057, so the flattening is a design decision made inside the framework rather than an automatic property of it, though the return of the distortion in this run is smaller than the framework predicts.

The symbols are read as measurement, with a caveat. Test-time channel surgery on the generalized model shows the model leaning on the symbols: zeroing them collapses macro AUC to 0.608 and rare recall to 0.023, and zeroing RGB (symbols only) still reaches 0.527 AUC. The shortcut test shuffles the symbol channels across test images, destroying their alignment with the underlying picture while preserving their per-image statistics; AUC falls by 0.108 (0.796 to 0.688), so alignment carries real signal. The remaining gap between shuffled (0.688) and zeroed (0.608) indicates that some per-image symbol statistics are also being used, a partial shortcut component we report rather than suppress.

Table 3: Balanced AURC (lower is better) for five abstention signals, reported per run and never pooled. The two runs differ in training budget and seed protocol, so pooling would be invalid; the ranking reversal is the point.

signal	reduced budget run	full-data run (3 seeds)
softmax confidence	0.639	0.512
random ranking	0.605	0.517
deep ensemble disagreement	0.589	0.543
source certificate (label free)	0.618	0.545
MC-dropout uncertainty	0.658	0.562

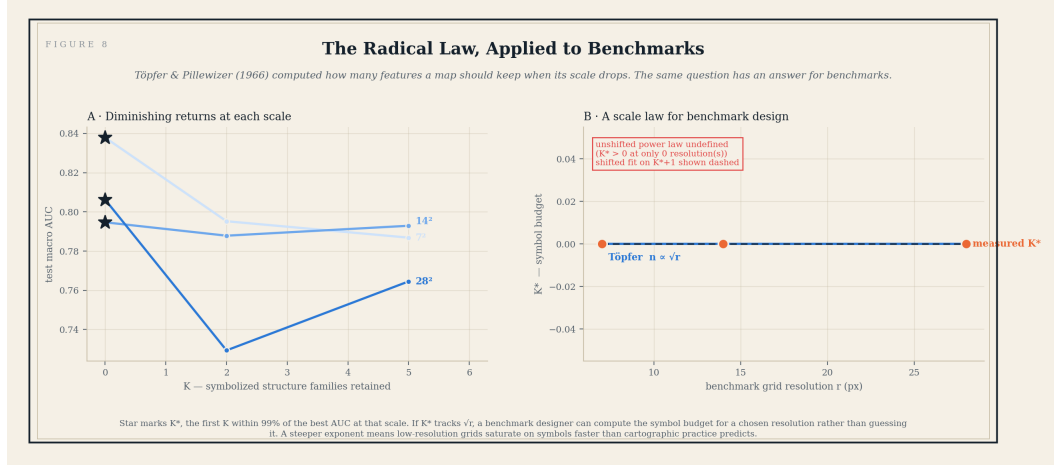


Figure 4: The radical law, tested. K^* marks the smallest symbol budget within 99% of the best AUC at each scale. Measured $K^* = 0$ at 7^2 , 14^2 , and 28^2 : no symbol family cleared the criterion at any resolution under this budget, and the predicted $\sqrt{r}$ law cannot be fit.

The symbols localize what experts annotate, and only that. Against ISIC 2018 Task 2 expert attribute masks (Codella et al., 2018) on 200 images (Figure discussion; per-attribute values printed in the released figure), within-image pixel AUC is 0.776 for pigment network, 0.672 for dots and globules, and 0.867 for streaks, while image-level AUC (does symbol density separate images with the attribute from images without) is near chance at 0.590, 0.465, and 0.484. Two of the five families (vessels, blue-white veil) have no ISIC counterpart and are reported as measured but unvalidated.

Negative result: the radical law does not appear. Töpfer’s radical law predicts that the number of features a map should keep scales as $\sqrt{r}$ with scale, which would let a benchmark designer compute a symbol budget for a chosen resolution. Sweeping resolution (7^2 , 14^2 , 28^2) against the number of symbolized families and marking K^* , the first budget within 99% of the best AUC at that scale, gives $K^* = 0$ at every resolution (Figure 4): under this training budget, no symbol family cleared the inclusion criterion at any scale, and at 28^2 adding two families lowers sweep AUC by roughly 0.07 before partial recovery at five. The power-law fit is undefined through zeros. The cartographic analogy holds structurally in Sections 3 and 4 and fails quantitatively here.

The certificate as an abstention signal: unstable across runs. The source diagram needs no label and no classifier, so it can ship with the dataset rather than be refitted per model. Scored as a selective prediction signal (Geifman and El-Yaniv, 2017) on balanced coverage-risk curves (Table 3), the certificate beats the model’s own softmax confidence on the reduced budget run (AURC 0.618 versus 0.639) and loses to random ranking (0.605) and to deep ensemble disagreement (Lakshminarayanan et al., 2017) (0.589). On the independent full-data run of Section 5 the ordering reverses: softmax confidence is best (0.512) and the certificate falls behind random (0.545 versus 0.517). Only two facts replicate across the runs: MC-dropout (Gal and Ghahramani, 2016) is worst in both, and no signal separates decisively from trivial baselines. We therefore report the certificate as a shippable artifact whose abstention value is unproven, and we treat the run-to-run reversal itself as a finding (Section 5).

Table 4: Full-data run (three seeds, no subsampling). The super-resolution arm trains the native arm’s ResNet-18 and carries a capacity confound; all other low resolution arms train the identical small CNN.

regime	macro AUC	balanced acc.	rare recall	ECE
naive resize 28^2	0.900	0.579	0.602	0.037
generalized 28^2	0.858	0.483	0.607	0.065
native 224^2 (ResNet-18)	0.927	0.681	0.717	0.031
super-resolution of the 28^2 product (ResNet-18)	0.918	0.624	0.631	0.039
learned linear pooling	0.897	0.566	0.567	0.051
learned attention pooling	0.897	0.578	0.594	0.050
pan-sharpening (Brovey)	0.902	0.590	0.612	0.038
pan-sharpening (IHS)	0.899	0.576	0.602	0.034

5 A second run moves the conclusions

An independently executed run of the same codebase on the full training split (three seeds, no subsampling, additional baseline arms) yields Table 4. Absolute numbers are not comparable to Table 2 and we never pool the two. Three things move. First, the naive-to-native gap widens (rare recall 0.602 to 0.717) and the generalized arm’s rare recall advantage over naive shrinks to noise (0.607 versus 0.602). Second, a classical super-resolution baseline (Lanczos upsampling of the shipped 28^2 image plus unsharp masking (Dong et al., 2014), fed to the same ResNet-18 as the native arm) reaches 0.918 macro AUC, close to the native 0.927, so a generic prior recovers most of the aggregate gap without any symbolization; the comparison to the small-CNN arms carries an architecture confound we state plainly. Third, same-architecture control baselines land at parity with naive: learned linear and attention pooling initialized at area-mean (0.897, 0.897) and classical remote sensing pan-sharpening by Brovey and IHS fusion (Gillespie et al., 1987; Chavez et al., 1991) (0.902, 0.899), which is evidence that the pooling rule is not the bottleneck and that the literal remote sensing analogue of symbolize-then-fuse does not transfer for free. Together with the abstention reversal of Table 3, the two runs bracket which claims are stable (the Nyquist audit, which is a property of the data; the retention asymmetry; the shortcut test) and which are run-dependent (aggregate orderings among low resolution arms; every abstention ranking).

6 Discussion: readiness as a shippable artifact

The audit results that matter for data readiness are the ones that need no classifier. The Nyquist audit (Table 1) is arithmetic on acquisition and grid geometry; it could have been printed in the original release and would have told every downstream user that pigment network, dots, streaks, and vessels are unrepresentable on the shipped grid. The source diagram (Figure 1) is computable per image at test time with no labels, so it can be distributed with the data rather than refitted per model. The retention analysis (spread 0.094 versus 0.045) is a property of the reduction policy, measured with a probe rather than a task model. All three survive unchanged across our two runs, while model-dependent conclusions (aggregate orderings, every abstention ranking) do not; a readiness standard should therefore privilege exactly the artifacts that are properties of the data. The proposal above is that standard, and the machine-readable legend plus pinned invariants make it auditable by tooling rather than by trust. **What a benchmark should ship.** Every result above is computable for any MedMNIST-style dataset today. We propose that a downsampled medical imaging benchmark ship four artifacts with the data: (1) named reduction operators with fidelity contracts, each contract carrying a machine-checkable invariant; (2) a scale bar in physical units, so a reader can decide whether a structure is representable before training anything; (3) a source diagram, the per-cell certificate of where the acquisition supports measurement, published with the images since it needs no labels; and (4) a reliability diagram with ECE before and after calibration. The audit says where the map may be trusted; it does not by itself make the map better, as our own aggregate numbers show.

Limitations. One dataset, one modality, and two runs. The main tables come from the released pipeline’s reduced budget configuration and the full-data run uses a different budget, so between-run comparisons are qualitative by design. ITA estimated from contact-illuminated dermoscopy

is a stratification variable rather than colorimetric ground truth, and the test population is heavily skewed light. Two of five symbol families are unvalidated against expert annotation. The released configuration documents that the 18 channel arm needs substantially more optimization steps than the 3 channel arm to escape a chance-level plateau, so the generalized arm’s aggregate deficit may partly reflect budget sensitivity rather than representation; we report it as measured.

Use of large language models. Large language models were used as coding and writing assistants for the analysis pipeline and for drafting and editing this manuscript under author direction. All experiments, numbers, and figures come from the released code; the authors verified every reported value against the run outputs and take full responsibility for the content.

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